



NTI Retail Sales Analytics Project

A comprehensive business intelligence initiative transforming multichannel retail data into executive-ready insights.



Project Overview

Dataset

Multichannel retail sales system
covering 2023–2025 fiscal
periods

Scope

5,000 transactions · 5 stores ·
200 customers · 50 products

Deliverables

Python · SQL · Power BI · Excel KPI models · Tableau interactive stories

Team & Technical Stack



Rahma Omar —
Python

Data preprocessing,
cleaning, and
dimensional
modeling (Star
Schema)



Magdy
Elhosseny —
SQL
Customer behavior,
product profitability,
and regional
performance analysis



Tasneem Magdy
— Power BI
Data model, DAX
measures, Home &
Sales Overview
pages



Habiba Hany —
Power BI
Product & Customer
pages, theme,
formatting, and
navigation



Ahmed Maged
— Excel
Data transformation,
Power Query, and
table relationships

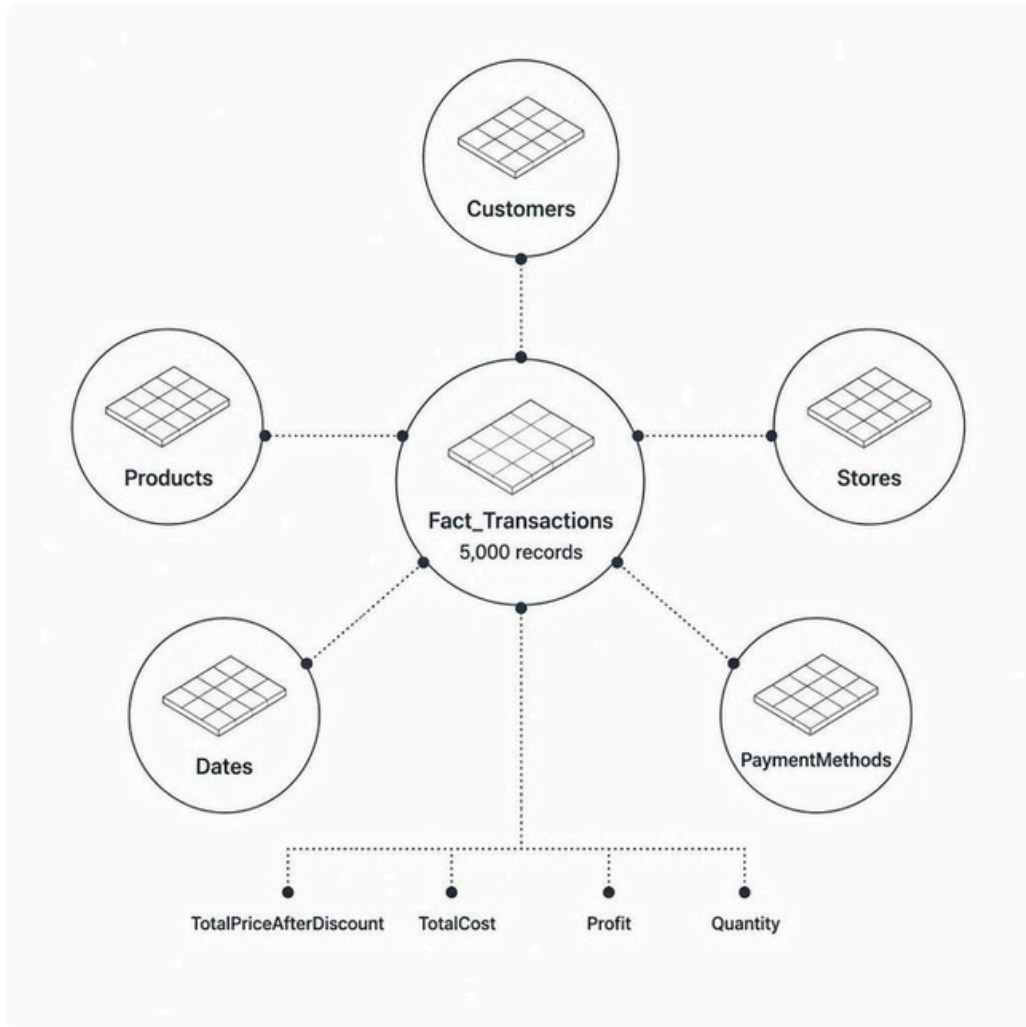


Mariam Mahran
— Excel
KPI modeling, pivot
tables, and
interactive dashboard



Mohamed Nabil
— Tableau
Interactive
dashboards,
storyboard, and KPI
cards

Data Architecture: Star Schema



Fact Table

Fact_Transactions — 5,000 records capturing all financial metrics at the transaction level.

Dimension Tables

- Customers
- Products
- Stores
- Dates
- PaymentMethods

Key Metrics Tracked

- TotalPriceAfterDiscount
- TotalCost & Profit
- Quantity

Executive KPIs

\$14.3M

TotalSales

Afterdiscount – \$14,301,904

\$3.8M

Total Profit

Netprofitof \$3,826,315

\$10.5M

TotalCost

Operational cost \$10,475,589

\$1.17M

TotalDiscounts

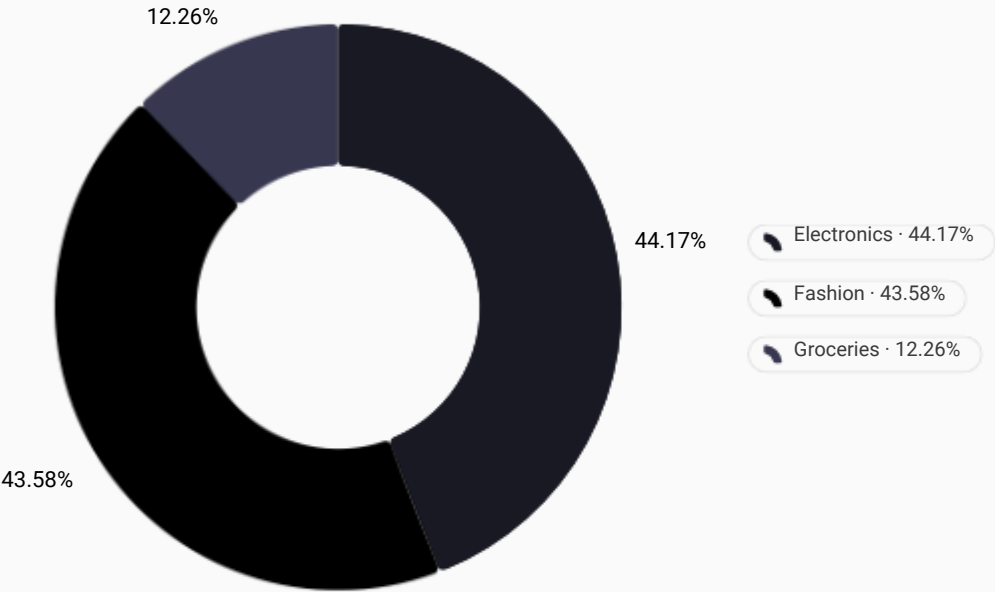
Discountamount \$1,173,269

5,000

Transactions

14,949total units sold

Sales Performance by Category



Category Breakdown

Revenue is nearly evenly split between two dominant categories, with Groceries serving a specialized niche.

Electronics — 44.2%

\$6,316,918 · Toprevenue driver

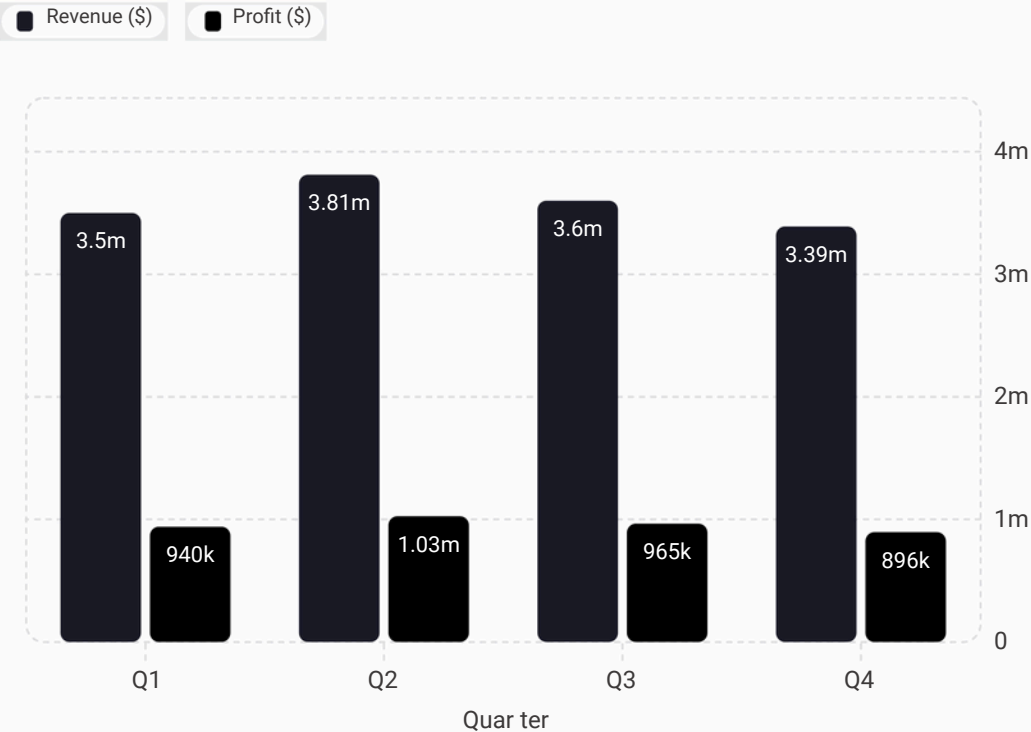
Fashion — 43.6%

\$6,232,280 · Close second

Groceries — 12.2%

\$1,752,706 · High-frequency, lower-margin

Regional & Quarterly Insights



Top Region

East leads all regions with **\$5,600,681** in revenue — **39.2%** of total sales.

Peak Quarter

Q2 achieved peak performance: **\$3,812,135** revenue and **\$1,025,527** profit.

Payment Leaders

- Cash — 1,284 transactions
- Credit Card — 1,281 transactions
- Mobile Money — 1,276 transactions

Analytics Deliverables



Power BI

4-page interactive dashboard — Home, Sales Overview, Product, Customer — with KPI cards, trend charts, regional analysis, and dynamic slicers



Excel

KPI score card, 4 pivot tables, and a multi-dimensional dashboard with cross-filtering capabilities



Tableau

3-part story— Overview, Analysis, Storyboard — with store performance and sub-category profitability views

Key Achievements



- 1 Scalable Star Schema**
Built a unified data architecture supporting 7 analytical tools across Python, SQL, Power BI, Excel, and Tableau
- 2 360° Business Intelligence**
Delivered comprehensive coverage of customer behavior, product profitability, regional performance, and temporal trends
- 3 Unified Design Language**
Consistent KPI styling, navigation, and visual hierarchy maintained across all dashboards and platforms